CNNP SKILL 2

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [195 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [122 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [157 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [602 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [211 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [70.7 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [62.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [125 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [583 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [21.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [4168 B]
Get:22 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3136 B]
Get:23 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:24 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [121 kB]
Get:25 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [20.0 kB]
Get:26 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Fetched 5045 kB in 3s (1586 kB/s)
189 packages can be upgraded. Run 'apt list --upgradable' to see them.
nishanth@LAPTOP-6Q3ONO7H:~$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
netcat-openbsd is already the newest version (1.234-1).
netcat-openbsd set to manually installed.
Upgrading:
  iproute2  libpcre2-8-0  zlib1g
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
netcat-openbsd is already the newest version (1.234-1).
netcat-openbsd set to manually installed.
Upgrading:
  iproute2  libpcre2-8-0  zlib1g

Installing:
  iperf3  mtr-tiny  tshark  wireshark

Installing dependencies:
  gnome-themes-extra-data  libdav1d7             libinput10        libopencore-amrnb0  libqt6opengl6          librist4          libtbb12          libx264-165          libzmq5
  ibverbs-providers        libdouble-conversion3 libiperf0         libopengl0          libqt6printsupport6    libsbc1           libtbbbind-2-5    libx265-215          libzvbi-common
  libaacs0                 libdvdnav4            libjsoncpp26      libopenjp2-7        libqt6qml6             libsctp1          libtbbmalloc2     libxcb-cursor0       libzvbi0t64
  libabsl20260107          libdvdread8t64        liblua5.4-0       libopenmpt0t64      libqt6qmlmeta6         libshine3         libts0t64         libxcb-icccm4        ocl-icd-libopencl1
  libavcodec62             libevdev2             libmaxminddb0     libpcap0.8t64       libqt6qmlmodels6       libsm6            libudfread3       libxcb-image0        qt6-gtk-platformtheme
  libavformat62            libfalcosecurity0t64  libmbedcrypto16   libpcre2-16-0       libqt6qmlworkerscript6 libsmi2t64        libva-x11-2       libxcb-keysyms1      qt6-qpa-plugins
  libavutil60              libgme0               libmd4c0          libpgm-5.3-0t64     libqt6quick6           libsnappy1v5      libvdpau1         libxcb-render-util0  qt6-svg-plugins
  libb2-1                  libgrpc++1.51t64      libminizip1t64    libprotobuf32t64    libqt6svg6             libsoxr0          libvorbisfile3    libxcb-shape0        qt6-translations-l10n
  libbcg729-0              libgrpc29t64          libmtdev1t64      libprotoc32t64      libqt6waylandclient6   libspandsp2t64    libvpl2           libxcb-util1         qt6-wayland
  libbdplus0               libgsm1               libnghttp3-9      libqt6core5compat6  libqt6waylandcompositor6 libspeexdsp1    libwacom-common   libxcb-xinput0       wireshark-common
  libbluray3               libhwloc-plugins      libnl-3-200       libqt6core6t64      libqt6widgets6         libsrt1.5-gnutls  libwacom9         libxcb-xkb1
  libcares2                libhwloc15            libnl-gen1-3-200  libqt6dbus6         libqt6wlshellintegration6 libssh-4        libwireshark-data libxkbcommon-x11-0
  libchromaprint1          libibverbs1           libnl-route-3-200 libqt6gui6          librabbitmq4           libsvtav1enc2     libwireshark19    libxnvctrl0
  libcjson1                libice6               libnorm1t64       libqt6multimedia6   librav1e0.8            libswresample6    libwiretap16      libxvidcore4
  libcodec2-1.2            libinput-bin          libnuma1          libqt6network6      libre2-11              libswscale9       libwsutil17       libze1

Suggested packages:
  libcuda1     libnvidia-encode1  libdvdcss2               mmdb-bin              lksctp-tools         libwacom-bin  geoip-database       libjs-leaflet              wireshark-doc
  libnvcuvid1  libbluray-bdj      libhwloc-contrib-plugins qt6-qmltooling-plugins snmp-mibs-downloader geoipupdate  geoip-database-extra libjs-leaflet.markercluster opencl-icd

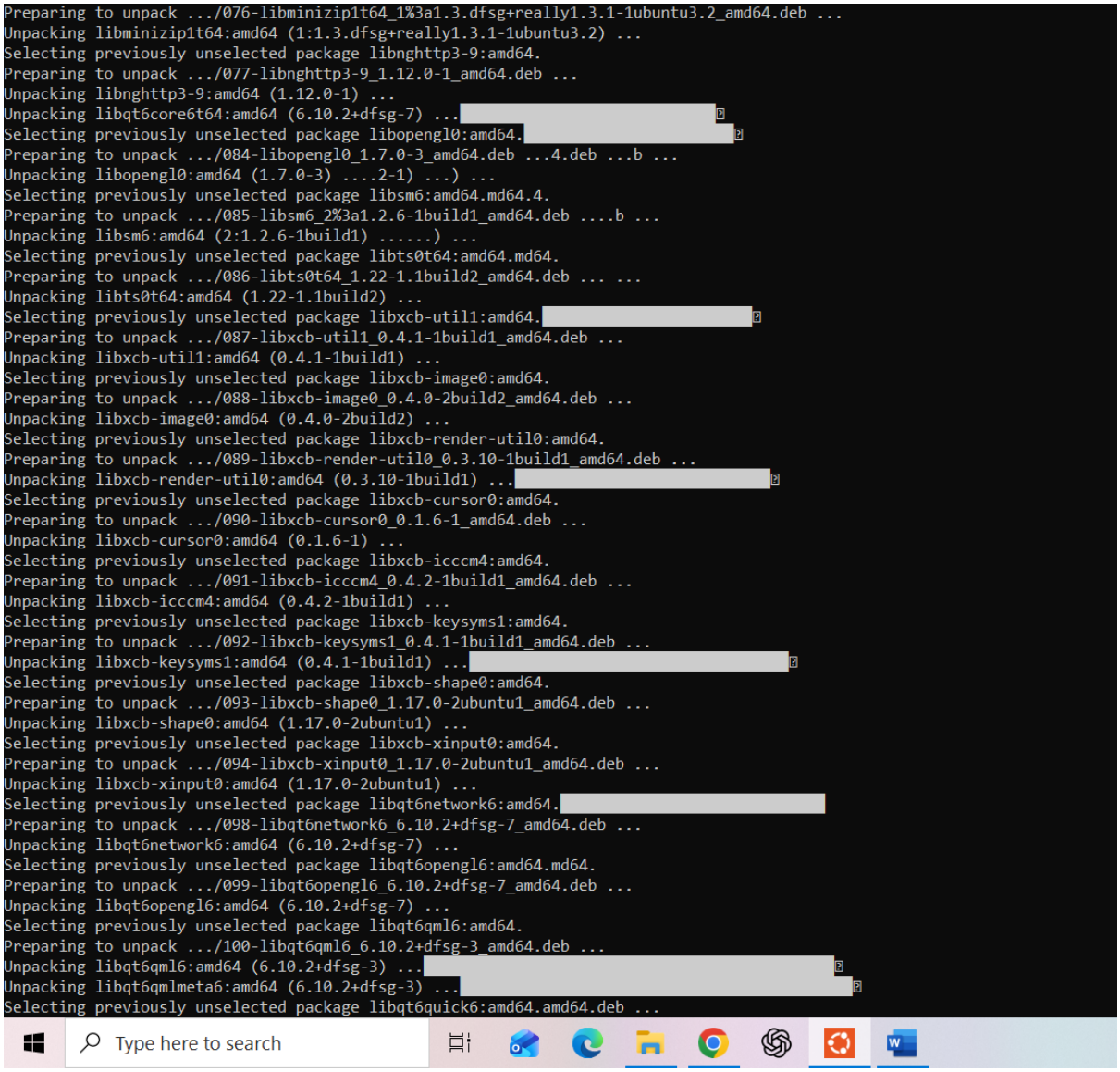
Summary:
  Upgrading: 3, Installing: 134, Removing: 0, Not Upgrading: 186
  Download size: 95.8 MB
  Space needed: 375 MB / 1021 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsctp1 amd64 1.0.21+dfsg-1build2 [9352 B]
```

```
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsctp1 amd64 1.0.21+dfsg-1build2 [9352 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libiperf0 amd64 3.20-2 [93.2 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 iperf3 amd64 3.20-2 [19.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpcre2-8-0 amd64 10.46-1ubuntu0.1 [257 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 zlib1g amd64 1:1.3.dfsg+really1.3.1-1ubuntu3.2 [61.8 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-1ubuntu1.1 [1198 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnl-3-200 amd64 3.12.0-2 [58.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnl-route-3-200 amd64 3.12.0-2 [204 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 libibverbs1 amd64 61.0-2ubuntu3 [70.6 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 ibverbs-providers amd64 61.0-2ubuntu3 [417 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 libevdev2 amd64 1.13.6+dfsg-1 [37.7 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmaxminddb0 amd64 1.12.2-1build2 [18.9 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnuma1 amd64 2.0.19-1build1 [23.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpcap0.8t64 amd64 1.10.6-1ubuntu1.1 [160 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 mtr-tiny amd64 0.95-1.1ubuntu2 [54.3 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 gnome-themes-extra-data all 3.28-5 [69.4 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libaacs0 amd64 0.11.1-4build1 [60.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute/main amd64 libabsl20260107 amd64 20260107.0-4 [588 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libva-x11-2 amd64 2.23.0-1ubuntu1 [12.2 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/main amd64 libvdpau1 amd64 1.5-4 [27.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libvpl2 amd64 1:2.16.0-1 [134 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute/main amd64 ocl-icd-libopencl1 amd64 2.3.4-1 [40.9 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libavutil60 amd64 7:8.0.1-3ubuntu2 [449 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libcodec2-1.2 amd64 1.2.0-4 [9015 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libdav1d7 amd64 1.5.3-1 [626 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libgsm1 amd64 1.0.23-2 [28.9 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libopenjp2-7 amd64 2.5.4-1ubuntu0.1 [187 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute/universe amd64 librav1e0.8 amd64 0.8.1-7 [1062 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libshine3 amd64 3.1.1-3build1 [23.5 kB]
Get:30 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsnappy1v5 amd64 1.2.2-2 [30.8 kB]
Get:31 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libsvtav1enc2 amd64 2.3.0+dfsg-1build1 [2742 kB]
Get:32 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libsoxr0 amd64 0.1.3-4.1 [82.7 kB]
Get:33 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libswresample6 amd64 7:8.0.1-3ubuntu2 [77.0 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libx264-165 amd64 2:0.165.3222+gitb35605ac-3build1 [609 kB]
Get:35 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libx265-215 amd64 4.1-4 [1331 kB]
Get:36 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libxvidcore4 amd64 2:1.3.7-3 [220 kB]
Get:37 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libzvbi-common all 0.2.44-1ubuntu2 [43.0 kB]
Get:38 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libzvbi0t64 amd64 0.2.44-1ubuntu2 [264 kB]
Get:39 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libavcodec62 amd64 7:8.0.1-3ubuntu2 [9711 kB]
Get:40 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libudfread3 amd64 1.2.0-2 [17.3 kB]
Get:41 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libbluray3 amd64 1:1.4.1-1 [133 kB]
Get:42 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libchromaprint1 amd64 1.6.0-2build1 [33.4 kB]
Get:43 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libdvdread8t64 amd64 7.0.1-1 [63.3 kB]
Get:44 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libdvdnav4 amd64 7.0.0-2 [40.3 kB]
Get:45 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libgme0 amd64 0.6.4-1 [146 kB]
Get:46 http://archive.ubuntu.com/ubuntu resolute/main amd64 libvorbisfile3 amd64 1.3.7-3build2 [17.2 kB]
```

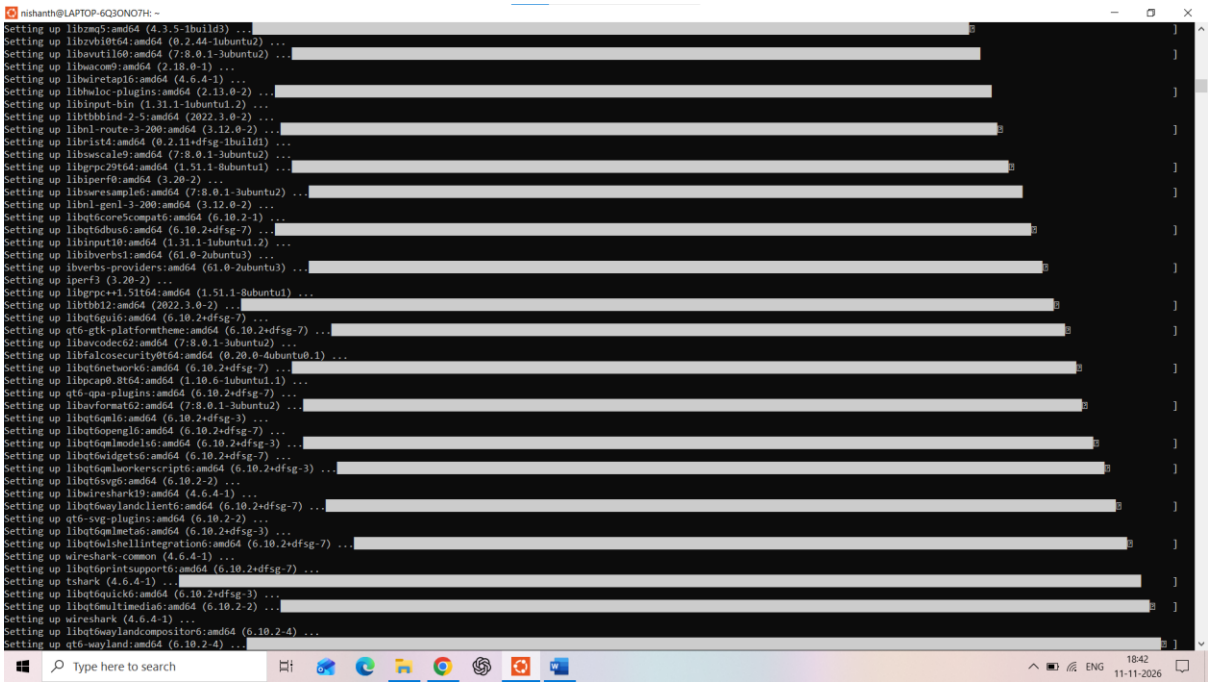
```
Get:70 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libtbbbind-2-5 amd64 2022.3.0-2 [15.4 kB]
Get:71 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libtbbmalloc2 amd64 2022.3.0-2 [54.0 kB]
Get:72 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libtbb12 amd64 2022.3.0-2 [98.2 kB]
Get:73 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 libfalcosecurity0t64 amd64 0.20.0-4ubuntu0.1 [1673 kB]
Get:74 http://archive.ubuntu.com/ubuntu resolute/main amd64 libice6 amd64 2:1.1.1-1build1 [44.0 kB]
Get:75 http://archive.ubuntu.com/ubuntu resolute/main amd64 libwacom-common all 2.18.0-1 [117 kB]
Get:76 http://archive.ubuntu.com/ubuntu resolute/main amd64 libwacom9 amd64 2.18.0-1 [28.4 kB]
Get:77 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libinput-bin amd64 1.31.1-1ubuntu1.2 [24.5 kB]
Get:78 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmtdev1t64 amd64 1.1.7-1build1 [15.2 kB]
Get:79 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libinput10 amd64 1.31.1-1ubuntu1.2 [161 kB]
Get:80 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblua5.4-0 amd64 5.4.8-1build1 [163 kB]
Get:81 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libmd4c0 amd64 0.5.2-2build1 [43.7 kB]
Get:82 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 libminizip1t64 amd64 1:1.3.dfsg+really1.3.1-1ubuntu3.2 [22.
Get:83 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libnghttp3-9 amd64 1.12.0-1 [63.6 kB]
Get:84 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnl-genl-3-200 amd64 3.12.0-2 [12.4 kB]
Get:85 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libopencore-amrnb0 amd64 0.1.6-1build2 [101 kB]
Get:86 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpcre2-16-0 amd64 10.46-1ubuntu0.1 [241 kB]
Get:87 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6core6t64 amd64 6.10.2+dfsg-7 [2080 kB]
Get:88 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6core5compat6 amd64 6.10.2-1 [150 kB]
Get:89 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6dbus6 amd64 6.10.2+dfsg-7 [285 kB]
Get:90 http://archive.ubuntu.com/ubuntu resolute/main amd64 libopengl0 amd64 1.7.0-3 [32.8 kB]
Get:91 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsm6 amd64 2:1.2.6-1build1 [16.9 kB]
Get:92 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libts0t64 amd64 1.22-1.1build2 [64.0 kB]
Get:93 http://archive.ubuntu.com/ubuntu resolute/main amd64 libxcb-util1 amd64 0.4.1-1build1 [11.1 kB]
Get:94 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libxcb-image0 amd64 0.4.0-2build2 [11.0 kB]
Get:95 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libxcb-render-util0 amd64 0.3.10-1build1 [10.4 kB]
Get:96 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libxcb-cursor0 amd64 0.1.6-1 [10.6 kB]
Get:97 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libxcb-icccm4 amd64 0.4.2-1build1 [11.0 kB]
Get:98 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libxcb-keysyms1 amd64 0.4.1-1build1 [8704 B]
Get:99 http://archive.ubuntu.com/ubuntu resolute/main amd64 libxcb-shape0 amd64 1.17.0-2ubuntu1 [6172 B]
Get:100 http://archive.ubuntu.com/ubuntu resolute/main amd64 libxcb-xinput0 amd64 1.17.0-2ubuntu1 [32.5 kB]
Get:101 http://archive.ubuntu.com/ubuntu resolute/main amd64 libxcb-xkb1 amd64 1.17.0-2ubuntu1 [32.7 kB]
Get:102 http://archive.ubuntu.com/ubuntu resolute/main amd64 libxkbcommon-x11-0 amd64 1.13.1-1 [15.0 kB]
Get:103 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6gui6 amd64 6.10.2+dfsg-7 [3550 kB]
Get:104 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6network6 amd64 6.10.2+dfsg-7 [859 kB]
Get:105 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6opengl6 amd64 6.10.2+dfsg-7 [439 kB]
Get:106 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6qml6 amd64 6.10.2+dfsg-3 [1893 kB]
Get:107 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6qmlmodels6 amd64 6.10.2+dfsg-3 [387 kB]
Get:108 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6qmlworkerscript6 amd64 6.10.2+dfsg-3 [35.9 kB]
Get:109 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6qmlmeta6 amd64 6.10.2+dfsg-3 [69.4 kB]
Get:110 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6quick6 amd64 6.10.2+dfsg-3 [2514 kB]
Get:111 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libswscale9 amd64 7:8.0.1-3ubuntu2 [411 kB]
Get:112 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6multimedia6 amd64 6.10.2-2 [831 kB]
Get:113 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6widgets6 amd64 6.10.2+dfsg-7 [2818 kB]
Get:114 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6printsupport6 amd64 6.10.2+dfsg-7 [224 kB]
Get:115 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libqt6svg6 amd64 6.10.2-2 [234 kB]
```

```
Preparing to unpack .../050-libzmq5_4.3.5-1build3_amd64.deb ...
Unpacking libzmq5:amd64 (4.3.5-1build3) ...
Selecting previously unselected package libavformat62:amd64.
Preparing to unpack .../051-libavformat62_7%3a8.0.1-3ubuntu2_amd64.deb ...
Unpacking libavformat62:amd64 (7:8.0.1-3ubuntu2) ...
Selecting previously unselected package libb2-1:amd64.
Preparing to unpack .../052-libb2-1_0.98.1-1.1build2_amd64.deb ...
Unpacking libb2-1:amd64 (0.98.1-1.1build2) ...
Selecting previously unselected package libbcg729-0:amd64.
Preparing to unpack .../053-libbcg729-0_1.1.1-3build1_amd64.deb ...
Unpacking libbcg729-0:amd64 (1.1.1-3build1) ...
Selecting previously unselected package libbdplus0:amd64.
Preparing to unpack .../054-libbdplus0_0.2.0-4build1_amd64.deb ...
Unpacking libbdplus0:amd64 (0.2.0-4build1) ...
Selecting previously unselected package libcares2:amd64.
Preparing to unpack .../055-libcares2_1.34.6-1ubuntu0.2_amd64.deb ...
Unpacking libcares2:amd64 (1.34.6-1ubuntu0.2) ...
Selecting previously unselected package libdouble-conversion3:amd64.
Preparing to unpack .../056-libdouble-conversion3_3.4.0-1_amd64.deb ...
Unpacking libdouble-conversion3:amd64 (3.4.0-1) ...
Selecting previously unselected package libprotobuf32t64:amd64.
Preparing to unpack .../057-libprotobuf32t64_3.21.12-15ubuntu1.1_amd64.deb ...
Unpacking libprotobuf32t64:amd64 (3.21.12-15ubuntu1.1) ...
Selecting previously unselected package libprotoc32t64:amd64.
Preparing to unpack .../058-libprotoc32t64_3.21.12-15ubuntu1.1_amd64.deb ...
Unpacking libprotoc32t64:amd64 (3.21.12-15ubuntu1.1) ...
Selecting previously unselected package libre2-11:amd64.
Preparing to unpack .../059-libre2-11_20250805-1build3_amd64.deb ...
Unpacking libre2-11:amd64 (20250805-1build3) ...
Selecting previously unselected package libgrpc29t64:amd64.
Preparing to unpack .../060-libgrpc29t64_1.51.1-8ubuntu1_amd64.deb ...
Unpacking libgrpc29t64:amd64 (1.51.1-8ubuntu1) ...
Selecting previously unselected package libgrpc++1.51t64:amd64.
Preparing to unpack .../061-libgrpc++1.51t64_1.51.1-8ubuntu1_amd64.deb ...
Unpacking libgrpc++1.51t64:amd64 (1.51.1-8ubuntu1) ...
Selecting previously unselected package libjsoncpp26:amd64.
Preparing to unpack .../062-libjsoncpp26_1.9.6-5_amd64.deb ...
Unpacking libjsoncpp26:amd64 (1.9.6-5) ...
Selecting previously unselected package libhwloc15:amd64.
Preparing to unpack .../063-libhwloc15_2.13.0-2_amd64.deb ...
Unpacking libhwloc15:amd64 (2.13.0-2) ...
Selecting previously unselected package libtbbbind-2-5:amd64.
Preparing to unpack .../064-libtbbbind-2-5_2022.3.0-2_amd64.deb ...
Unpacking libtbbbind-2-5:amd64 (2022.3.0-2) ...
Selecting previously unselected package libtbbmalloc2:amd64.
Preparing to unpack .../065-libtbbmalloc2_2022.3.0-2_amd64.deb ...
Unpacking libtbbmalloc2:amd64 (2022.3.0-2) ...
Selecting previously unselected package libtbb12:amd64.
Preparing to unpack .../066-libtbb12_2022.3.0-2_amd64.deb ...
Unpacking libtbb12:amd64 (2022.3.0-2) ...
```


Preparing to unpack .../076-libminizip1t64_1%3a1.3.dfsg+really1.3.1-1ubuntu3.2_amd64.deb ...
Unpacking libminizip1t64:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3.2) ...
Selecting previously unselected package libnghttp3-9:amd64.
Preparing to unpack .../077-libnghttp3-9_1.12.0-1_amd64.deb ...
Unpacking libnghttp3-9:amd64 (1.12.0-1) ...
Unpacking libqt6core6t64:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package libopengl0:amd64.
Preparing to unpack .../084-libopengl0_1.7.0-3_amd64.deb ...4.deb ...b ...
Unpacking libopengl0:amd64 (1.7.0-3) ....2-1) ...) ...
Selecting previously unselected package libsm6:amd64.md64.4.
Preparing to unpack .../085-libsm6_2%3a1.2.6-1build1_amd64.deb ....b ...
Unpacking libsm6:amd64 (2:1.2.6-1build1) ......) ...
Selecting previously unselected package libts0t64:amd64.md64.
Preparing to unpack .../086-libts0t64_1.22-1.1build2_amd64.deb ... ...
Unpacking libts0t64:amd64 (1.22-1.1build2) ...
Selecting previously unselected package libxcb-util1:amd64.
Preparing to unpack .../087-libxcb-util1_0.4.1-1build1_amd64.deb ...
Unpacking libxcb-util1:amd64 (0.4.1-1build1) ...
Selecting previously unselected package libxcb-image0:amd64.
Preparing to unpack .../088-libxcb-image0_0.4.0-2build2_amd64.deb ...
Unpacking libxcb-image0:amd64 (0.4.0-2build2) ...
Selecting previously unselected package libxcb-render-util0:amd64.
Preparing to unpack .../089-libxcb-render-util0_0.3.10-1build1_amd64.deb ...
Unpacking libxcb-render-util0:amd64 (0.3.10-1build1) ...
Selecting previously unselected package libxcb-cursor0:amd64.
Preparing to unpack .../090-libxcb-cursor0_0.1.6-1_amd64.deb ...
Unpacking libxcb-cursor0:amd64 (0.1.6-1) ...
Selecting previously unselected package libxcb-icccm4:amd64.
Preparing to unpack .../091-libxcb-icccm4_0.4.2-1build1_amd64.deb ...
Unpacking libxcb-icccm4:amd64 (0.4.2-1build1) ...
Selecting previously unselected package libxcb-keysyms1:amd64.
Preparing to unpack .../092-libxcb-keysyms1_0.4.1-1build1_amd64.deb ...
Unpacking libxcb-keysyms1:amd64 (0.4.1-1build1) ...
Selecting previously unselected package libxcb-shape0:amd64.
Preparing to unpack .../093-libxcb-shape0_1.17.0-2ubuntu1_amd64.deb ...
Unpacking libxcb-shape0:amd64 (1.17.0-2ubuntu1) ...
Selecting previously unselected package libxcb-xinput0:amd64.
Preparing to unpack .../094-libxcb-xinput0_1.17.0-2ubuntu1_amd64.deb ...
Unpacking libxcb-xinput0:amd64 (1.17.0-2ubuntu1) ...
Selecting previously unselected package libqt6network6:amd64.
Preparing to unpack .../098-libqt6network6_6.10.2+dfsg-7_amd64.deb ...
Unpacking libqt6network6:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package libqt6opengl6:amd64.md64.
Preparing to unpack .../099-libqt6opengl6_6.10.2+dfsg-7_amd64.deb ...
Unpacking libqt6opengl6:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package libqt6qml6:amd64.
Preparing to unpack .../100-libqt6qml6_6.10.2+dfsg-3_amd64.deb ...
Unpacking libqt6qml6:amd64 (6.10.2+dfsg-3) ...
Unpacking libqt6qmlmeta6:amd64 (6.10.2+dfsg-3) ...
Selecting previously unselected package libqt6quick6:amd64.amd64.deb ...
Type here to search

```
Preparing to unpack .../115-libspeexdsp1_1.2.1-3build1_amd64.deb ...
Unpacking libspeexdsp1:amd64 (1.2.1-3build1) ...
Selecting previously unselected package libwireshark-data.
Preparing to unpack .../116-libwireshark-data_4.6.4-1_all.deb ...
Unpacking libwireshark-data (4.6.4-1) ...
Selecting previously unselected package libsbc1:amd64.
Preparing to unpack .../117-libsbc1_2.1-1build1_amd64.deb ...
Unpacking libsbc1:amd64 (2.1-1build1) ...
Selecting previously unselected package libwsutil17:amd64.
Preparing to unpack .../118-libwsutil17_4.6.4-1_amd64.deb ...
Unpacking libwsutil17:amd64 (4.6.4-1) ...
Selecting previously unselected package libwiretap16:amd64.
Preparing to unpack .../119-libwiretap16_4.6.4-1_amd64.deb ...
Unpacking libwiretap16:amd64 (4.6.4-1) ...
Selecting previously unselected package libwireshark19:amd64.
Preparing to unpack .../120-libwireshark19_4.6.4-1_amd64.deb ...
Unpacking libwireshark19:amd64 (4.6.4-1) ...
Selecting previously unselected package libxnvctrl0:amd64.
Preparing to unpack .../121-libxnvctrl0_510.47.03-0ubuntu7_amd64.deb ...
Unpacking libxnvctrl0:amd64 (510.47.03-0ubuntu7) ...
Selecting previously unselected package libze1:amd64.
Preparing to unpack .../122-libze1_1.28.2-2_amd64.deb ...
Unpacking libze1:amd64 (1.28.2-2) ...
Selecting previously unselected package qt6-gtk-platformtheme:amd64.
Preparing to unpack .../123-qt6-gtk-platformtheme_6.10.2+dfsg-7_amd64.deb ...
Unpacking qt6-gtk-platformtheme:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package qt6-qpa-plugins:amd64.
Preparing to unpack .../124-qt6-qpa-plugins_6.10.2+dfsg-7_amd64.deb ...
Unpacking qt6-qpa-plugins:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package qt6-svg-plugins:amd64.
Preparing to unpack .../125-qt6-svg-plugins_6.10.2-2_amd64.deb ...
Unpacking qt6-svg-plugins:amd64 (6.10.2-2) ...
Selecting previously unselected package qt6-translations-l10n.
Preparing to unpack .../126-qt6-translations-l10n_6.10.2-1_all.deb ...
Unpacking qt6-translations-l10n (6.10.2-1) ...
Selecting previously unselected package qt6-wayland:amd64.
Preparing to unpack .../127-qt6-wayland_6.10.2-4_amd64.deb ...
Unpacking qt6-wayland:amd64 (6.10.2-4) ...
Selecting previously unselected package wireshark-common.
Preparing to unpack .../128-wireshark-common_4.6.4-1_amd64.deb ...
Unpacking wireshark-common (4.6.4-1) ...
Selecting previously unselected package tshark.
Preparing to unpack .../129-tshark_4.6.4-1_amd64.deb ...
Unpacking tshark (4.6.4-1) ...
Selecting previously unselected package wireshark.
Preparing to unpack .../130-wireshark_4.6.4-1_amd64.deb ...
Unpacking wireshark (4.6.4-1) ...
Selecting previously unselected package libhwloc-plugins:amd64.
Preparing to unpack .../131-libhwloc-plugins_2.13.0-2_amd64.deb ...
Unpacking libhwloc-plugins:amd64 (2.13.0-2) ...
```

```
Setting up libmaxminddb0:amd64 (1.12.2-1build2) ...
Setting up libts0t64:amd64 (1.22-1.1build2) ...
Setting up libxcb-keysyms1:amd64 (0.4.1-1build1) ...
Setting up libxcb-shape0:amd64 (1.17.0-2ubuntu1) ...
Setting up gnome-themes-extra-data (3.28-5) ...
Setting up libxcb-render-util0:amd64 (0.3.10-1build1) ...
Setting up libxcb-icccm4:amd64 (0.4.2-1build1) ...
Setting up libcjson1:amd64 (1.7.19-2) ...
Setting up libxvidcore4:amd64 (2:1.3.7-3) ...
Setting up libpcre2-16-0:amd64 (10.46-1ubuntu0.1) ...
Setting up libze1:amd64 (1.28.2-2) ...
Setting up libopengl0:amd64 (1.7.0-3) ...
Setting up libxcb-util1:amd64 (0.4.1-1build1) ...
Setting up libsnappy1v5:amd64 (1.2.2-2) ...
Setting up libxcb-xkb1:amd64 (1.17.0-2ubuntu1) ...
Setting up libxcb-image0:amd64 (0.4.0-2build2) ...
Setting up librav1e0.8:amd64 (0.8.1-7) ...
Setting up libxnvctrl0:amd64 (510.47.03-0ubuntu7) ...
Setting up qt6-translations-l10n (6.10.2-1) ...
Setting up libvorbisfile3:amd64 (1.3.7-3build2) ...
Setting up libpgm-5.3-0t64:amd64 (5.3.128~dfsg-2.1build2) ...
Setting up mtr-tiny (0.95-1.1ubuntu2) ...
Setting up libsvtav1enc2:amd64 (2.3.0+dfsg-1build1) ...
Setting up libjsoncpp26:amd64 (1.9.6-5) ...
Setting up libxcb-cursor0:amd64 (0.1.6-1) ...
Setting up libxkbcommon-x11-0:amd64 (1.13.1-1) ...
Setting up libhwloc15:amd64 (2.13.0-2) ...
Setting up libcares2:amd64 (1.34.6-1ubuntu0.2) ...
Setting up libaacs0:amd64 (0.11.1-4build1) ...
Setting up libssh-4:amd64 (0.11.3-1ubuntu2.1) ...
Setting up libnghttp3-9:amd64 (1.12.0-1) ...
Setting up libspandsp2t64:amd64 (0.0.6+dfsg-2.2build2) ...
Setting up libbdplus0:amd64 (0.2.0-4build1) ...
Setting up libnuma1:amd64 (2.0.19-1build1) ...
Setting up libsrt1.5-gnutls:amd64 (1.5.4-3) ...
Setting up libsctp1:amd64 (1.0.21+dfsg-1build2) ...
Setting up libwireshark-data (4.6.4-1) ...
Setting up libdav1d7:amd64 (1.5.3-1) ...
Setting up libmtdev1t64:amd64 (1.1.7-1build1) ...
Setting up libminizip1t64:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3.2) ...
Setting up ocl-icd-libopencl1:amd64 (2.3.4-1) ...
Setting up libvdpau1:amd64 (1.5-4) ...
Setting up libwsutil17:amd64 (4.6.4-1) ...
Setting up libspeexdsp1:amd64 (1.2.1-3build1) ...
Setting up libnl-3-200:amd64 (3.12.0-2) ...
Setting up libopencore-amrnb0:amd64 (0.1.6-1build2) ...
Setting up libmd4c0:amd64 (0.5.2-2build1) ...
Setting up liblua5.4-0:amd64 (5.4.8-1build1) ...
Setting up libopenjp2-7:amd64 (2.5.4-1ubuntu0.1) ...
Setting up libabsl20260107:amd64 (20260107.0-4) ...
```



Next, create the three network nodes:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns add host1
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns add router
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns add host2
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns list
host2
router
host1
nishanth@LAPTOP-6Q3ONO7H:~$
```

Create network cables & Connect cables to the three nodes:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip link add veth-h1 type veth peer name veth-r1
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip link add veth-r2 type veth peer name veth-h2
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip link set veth-h1 netns host1
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip link set veth-r1 netns router
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip link set veth-r2 netns router
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip link set veth-h2 netns host2
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
nishanth@LAPTOP-6Q3ONO7H:~$
```

Configure IP addresses:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ip link set veth-h1 up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ip link set lo up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip link set veth-r1 up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip link set veth-r2 up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip link set lo up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host2 ip link set veth-h2 up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host2 ip link set lo up
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo upError: ipv4: Address already assigned.
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo upError: ipv4: Address already assigned.
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo upError: ipv4: Address already assigned.
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
4: veth-h1@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether d6:7b:bc:63:eb:8a brd ff:ff:ff:ff:ff:ff link-netns router
    inet 10.0.1.1/30 scope global veth-h1
       valid_lft forever preferred_lft forever
```

```
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
4: veth-h1@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether d6:7b:bc:63:eb:8a brd ff:ff:ff:ff:ff:ff link-netns router
    inet 10.0.1.1/30 scope global veth-h1
       valid_lft forever preferred_lft forever
    inet6 fe80::d47b:bcff:fe63:eb8a/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
3: veth-r1@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 26:39:a8:4a:bf:7e brd ff:ff:ff:ff:ff:ff link-netns host1
    inet 10.0.1.2/30 scope global veth-r1
       valid_lft forever preferred_lft forever
    inet6 fe80::2439:a8ff:fe4a:bf7e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
6: veth-r2@if5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 02:9e:ab:1d:3c:4e brd ff:ff:ff:ff:ff:ff link-netns host2
    inet 10.0.2.1/30 scope global veth-r2
       valid_lft forever preferred_lft forever
    inet6 fe80::9e:abff:fe1d:3c4e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host2 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
5: veth-h2@if6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 9a:52:d5:04:0a:d3 brd ff:ff:ff:ff:ff:ff link-netns router
    inet 10.0.2.2/30 scope global veth-h2
       valid_lft forever preferred_lft forever
    inet6 fe80::9852:d5ff:fe04:ad3/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
nishanth@LAPTOP-6Q3ONO7H:~$
```

Enable routing:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
nishanth@LAPTOP-6Q3ONO7H:~$
```

Test the connection (PING):

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=5.05 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.138 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.096 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2032ms
rtt min/avg/max/mdev = 0.096/1.762/5.053/2.326 ms
nishanth@LAPTOP-6Q3ONO7H:~$
```

Configure network delay and packet loss:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 6775591488024920588
nishanth@LAPTOP-6Q3ONO7H:~$
```

Test ping again:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ping -c 5 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=20.6 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=20.5 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=20.6 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=20.2 ms
64 bytes from 10.0.2.2: icmp_seq=5 ttl=63 time=20.3 ms

--- 10.0.2.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 20.240/20.452/20.620/0.146 ms
```

In a new a terminal:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec host1 ping -c 10 10.0.2.2
[sudo: authenticate] Password:
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=20.5 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=20.3 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=20.6 ms
64 bytes from 10.0.2.2: icmp_seq=5 ttl=63 time=20.3 ms
64 bytes from 10.0.2.2: icmp_seq=6 ttl=63 time=20.5 ms
64 bytes from 10.0.2.2: icmp_seq=7 ttl=63 time=20.7 ms
64 bytes from 10.0.2.2: icmp_seq=8 ttl=63 time=20.4 ms
64 bytes from 10.0.2.2: icmp_seq=9 ttl=63 time=20.6 ms
64 bytes from 10.0.2.2: icmp_seq=10 ttl=63 time=20.4 ms

--- 10.0.2.2 ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 9042ms
rtt min/avg/max/mdev = 20.302/20.468/20.707/0.132 ms
nishanth@LAPTOP-6Q3ONO7H:~$
```

Verify the capture files:

Again in another terminal:

```
nishanth@LAPTOP-6Q3ONO7H:~$ ls -lh /tmp/cap*.pcap
-rw------- 1 root root 3.1K Nov 11 13:28 /tmp/cap1.pcap
-rw------- 1 root root 3.1K Nov 11 13:28 /tmp/cap2.pcap
-rw------- 1 root root 3.0K Nov 11 13:28 /tmp/cap3.pcap
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap1.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000 fe80::2439:a8ff:fe4a:bf7e → ff02::2      ICMPv6 70 Router Solicitation from 26:39:a8:4a:bf:7e
    2 110.516058103     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=1/256, ttl=64
    3 110.536506503     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=1/256, ttl=63 (request in 2)
    4 111.517119503     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=2/512, ttl=64
    5 112.543889603     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=3/768, ttl=64
    6 112.564161203     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=3/768, ttl=63 (request in 5)
    7 113.545764202     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=4/1024, ttl=64
    8 113.566264002     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=4/1024, ttl=63 (request in 7)
    9 114.548085900     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=5/1280, ttl=64
   10 114.568330300     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=5/1280, ttl=63 (request in 9)
   11 115.550104796     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=6/1536, ttl=64
   12 115.570549596     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=6/1536, ttl=63 (request in 11)
   13 115.711914296 d6:7b:bc:63:eb:8a → 26:39:a8:4a:bf:7e ARP 42 Who has 10.0.1.2? Tell 10.0.1.1
   14 115.711855196 26:39:a8:4a:bf:7e → d6:7b:bc:63:eb:8a ARP 42 Who has 10.0.1.1? Tell 10.0.1.2
   15 115.711980296 d6:7b:bc:63:eb:8a → 26:39:a8:4a:bf:7e ARP 42 10.0.1.1 is at d6:7b:bc:63:eb:8a
   16 115.712015896 26:39:a8:4a:bf:7e → d6:7b:bc:63:eb:8a ARP 42 10.0.1.2 is at 26:39:a8:4a:bf:7e
   17 116.552283692     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=7/1792, ttl=64
   18 116.572923692     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=7/1792, ttl=63 (request in 17)
   19 117.554380889     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=8/2048, ttl=64
   20 117.574673389     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=8/2048, ttl=63 (request in 19)
   21 118.556152785     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=9/2304, ttl=64
   22 118.576669985     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=9/2304, ttl=63 (request in 21)
   23 119.558130881     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=10/2560, ttl=64
   24 119.578417081     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=10/2560, ttl=63 (request in 23)
nishanth@LAPTOP-6Q3ONO7H:~$
```

Check Router packet capture:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap2.pcap
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=1/256, ttl=64
    2 0.020429300     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=1/256, ttl=63 (request in 1)
    3 1.001061600     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=2/512, ttl=64
    4 2.027834200     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=3/768, ttl=64
    5 2.048083400     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=3/768, ttl=63 (request in 4)
    6 3.029705299     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=4/1024, ttl=64
    7 3.050185799     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=4/1024, ttl=63 (request in 6)
    8 4.032028597     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=5/1280, ttl=64
    9 4.052252897     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=5/1280, ttl=63 (request in 8)
   10 5.034047093     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=6/1536, ttl=64
   11 5.054471293     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=6/1536, ttl=63 (request in 10)
   12 5.195768893 26:39:a8:4a:bf:7e → d6:7b:bc:63:eb:8a ARP 42 Who has 10.0.1.1? Tell 10.0.1.2
   13 5.195846993 d6:7b:bc:63:eb:8a → 26:39:a8:4a:bf:7e ARP 42 Who has 10.0.1.2? Tell 10.0.1.1
   14 5.195941893 26:39:a8:4a:bf:7e → d6:7b:bc:63:eb:8a ARP 42 10.0.1.2 is at 26:39:a8:4a:bf:7e
   15 5.195909893 d6:7b:bc:63:eb:8a → 26:39:a8:4a:bf:7e ARP 42 10.0.1.1 is at d6:7b:bc:63:eb:8a
   16 6.036224789     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=7/1792, ttl=64
   17 6.056845489     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=7/1792, ttl=63 (request in 16)
   18 7.038326586     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=8/2048, ttl=64
   19 7.058596386     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=8/2048, ttl=63 (request in 18)
   20 8.040097082     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=9/2304, ttl=64
   21 8.060592082     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=9/2304, ttl=63 (request in 20)
   22 9.042072878     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=10/2560, ttl=64
   23 9.062339978     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=10/2560, ttl=63 (request in 22)
```

Check Host 2 packet capture:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap3.pcap
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=1/256, ttl=63
    2 0.000293700     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=1/256, ttl=64 (request in 1)
    3 2.027873300     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=3/768, ttl=63
    4 2.027948200     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=3/768, ttl=64 (request in 3)
    5 3.029927899     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=4/1024, ttl=63
    6 3.030027799     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=4/1024, ttl=64 (request in 5)
    7 4.032049897     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=5/1280, ttl=63
    8 4.032119297     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=5/1280, ttl=64 (request in 7)
    9 5.034244893     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=6/1536, ttl=63
   10 5.034316493     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=6/1536, ttl=64 (request in 9)
   11 5.175695393 9a:52:d5:04:0a:d3 → 02:9e:ab:1d:3c:4e ARP 42 Who has 10.0.2.1? Tell 10.0.2.2
   12 5.196127392 02:9e:ab:1d:3c:4e → 9a:52:d5:04:0a:d3 ARP 42 Who has 10.0.2.2? Tell 10.0.2.1
   13 5.196174092 9a:52:d5:04:0a:d3 → 02:9e:ab:1d:3c:4e ARP 42 10.0.2.2 is at 9a:52:d5:04:0a:d3
   14 5.196129892 02:9e:ab:1d:3c:4e → 9a:52:d5:04:0a:d3 ARP 42 10.0.2.1 is at 02:9e:ab:1d:3c:4e
   15 6.036621789     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=7/1792, ttl=63
   16 6.036691589     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=7/1792, ttl=64 (request in 15)
   17 7.038384786     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=8/2048, ttl=63
   18 7.038465886     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=8/2048, ttl=64 (request in 17)
   19 8.040370782     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=9/2304, ttl=63
   20 8.040438682     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=9/2304, ttl=64 (request in 19)
   21 9.042105978     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=10/2560, ttl=63
   22 9.042185078     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=10/2560, ttl=64 (request in 21)
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap1.pcap -q -z io,stat,0
Running as user "root" and group "root". This could be dangerous.

===================================
| IO Statistics                   |
|                                 |
| Duration: 119.6 secs            |
| Interval: 119.6 secs            |
|                                 |
| Col  1: Frames and bytes        |
|---------------------------------|
|                |1               |
| Interval       | Frames | Bytes |
|---------------------------------|
|   0.0 <> 119.6 |     24 |  2100 |
===================================
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap2.pcap -q -z io,stat,0
Running as user "root" and group "root". This could be dangerous.

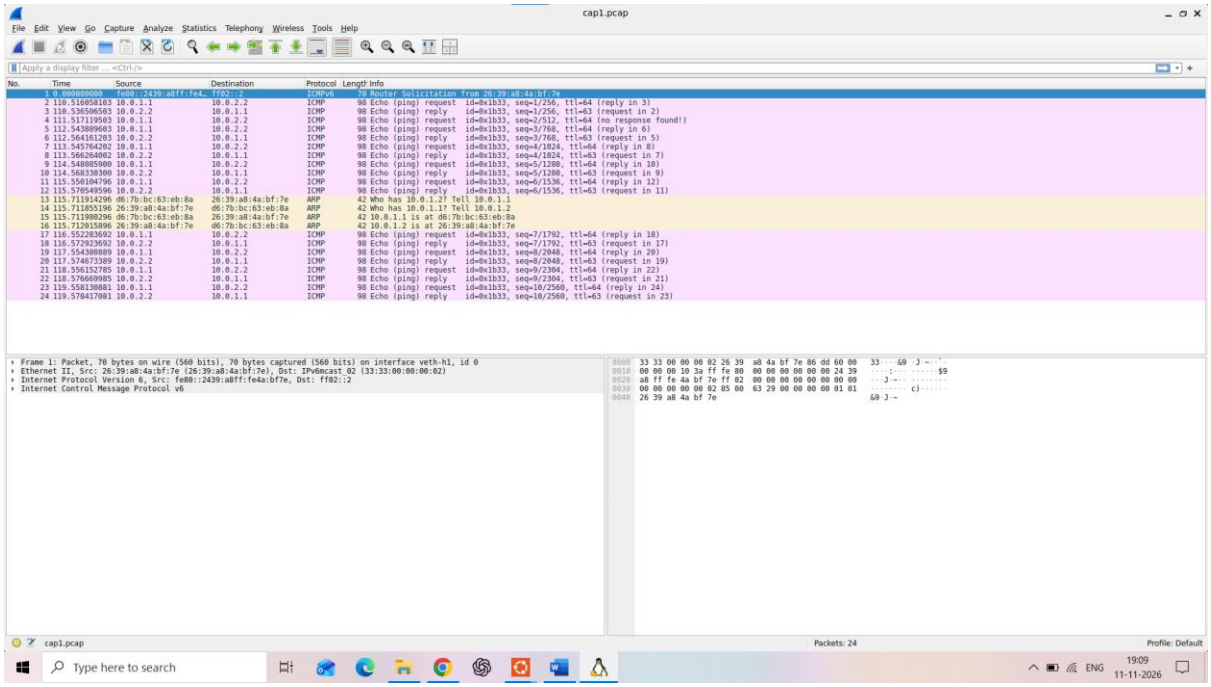
===================================
| IO Statistics                   |
|                                 |
| Duration: 9.062 secs            |
| Interval: 9.062 secs            |
|                                 |
| Col  1: Frames and bytes        |
|---------------------------------|
|                |1               |
| Interval       | Frames | Bytes |
|---------------------------------|
| 0.000 <> 9.062 |     23 |  2030 |
===================================
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap3.pcap -q -z io,stat,0
Running as user "root" and group "root". This could be dangerous.

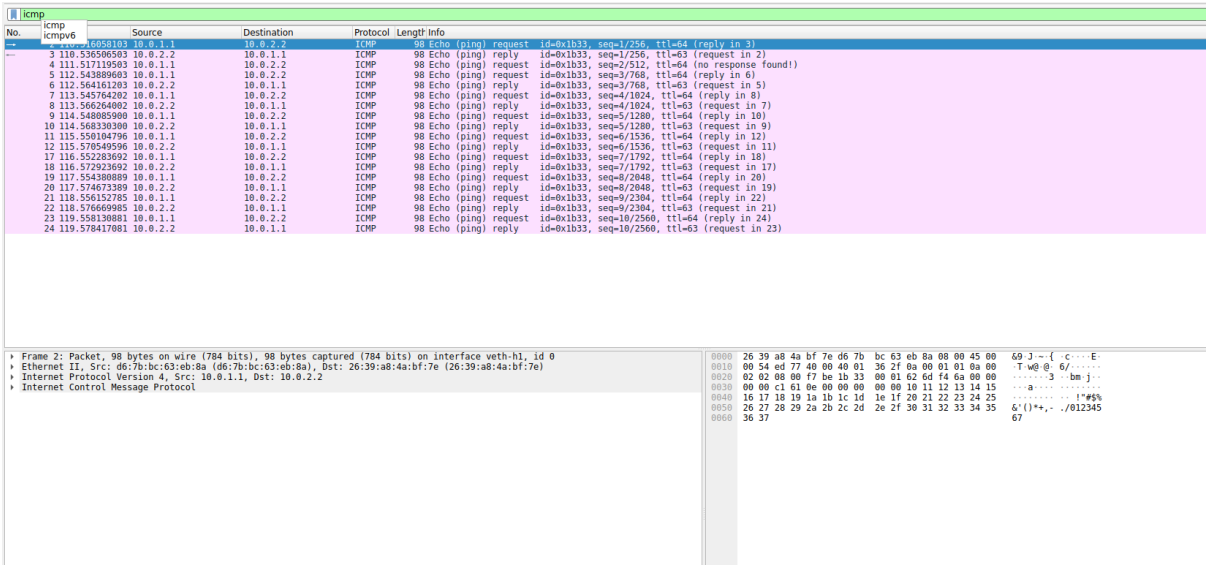
===================================
| IO Statistics                   |
|                                 |
| Duration: 9.042 secs            |
| Interval: 9.042 secs            |
|                                 |
| Col  1: Frames and bytes        |
|---------------------------------|
|                |1               |
| Interval       | Frames | Bytes |
|---------------------------------|
| 0.000 <> 9.042 |     22 |  1932 |
===================================
```

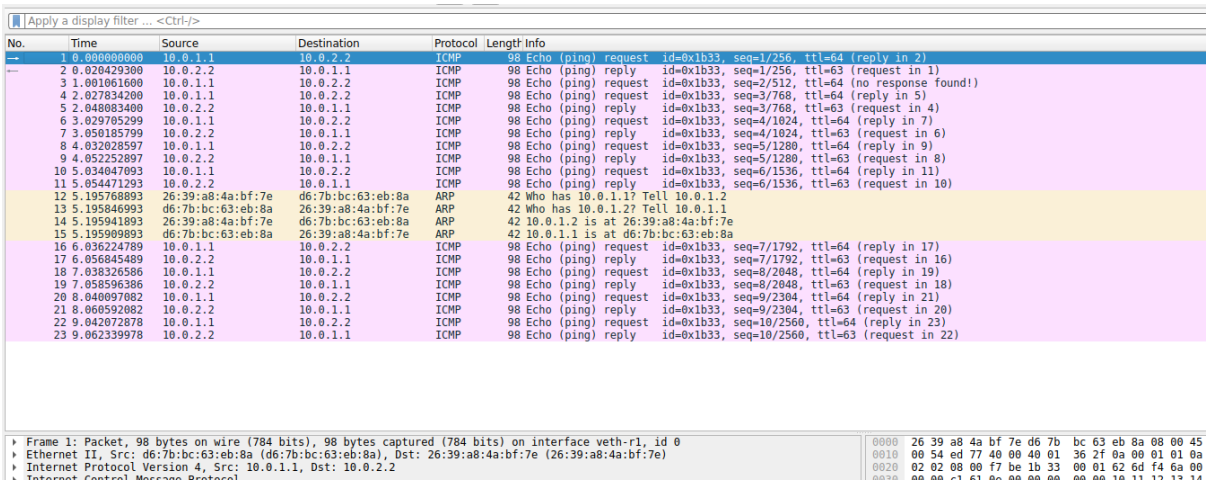
Analyze ICMP packets:

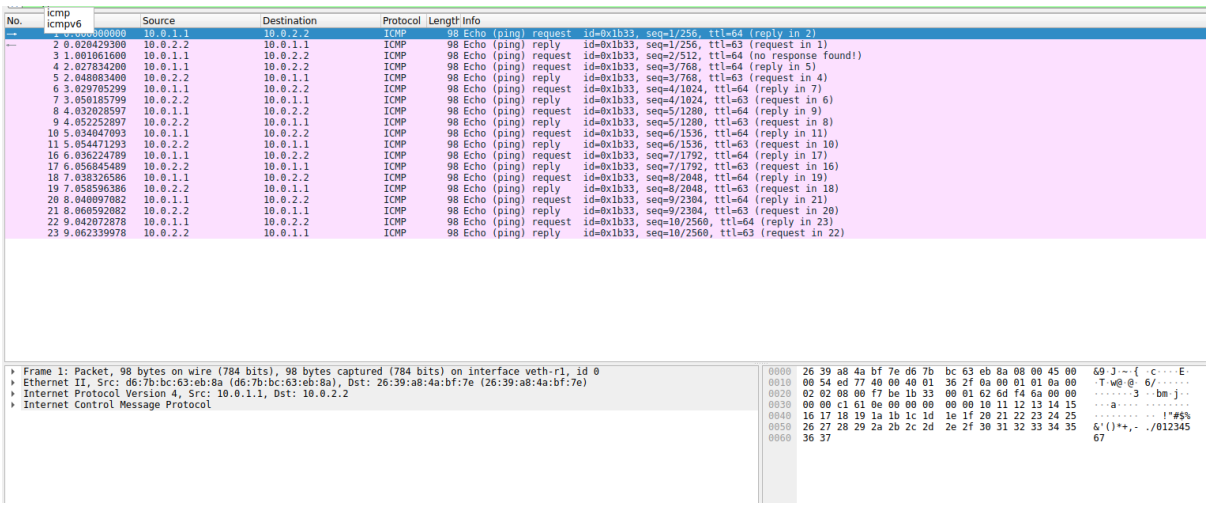
```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap1.pcap -Y icmp
Running as user "root" and group "root". This could be dangerous.
    2 110.516058103     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=1/256, ttl=64
    3 110.536506503     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=1/256, ttl=63 (request in 2)
    4 111.517119503     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=2/512, ttl=64
    5 112.543889603     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=3/768, ttl=64
    6 112.564161203     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=3/768, ttl=63 (request in 5)
    7 113.545764202     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=4/1024, ttl=64
    8 113.566264002     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=4/1024, ttl=63 (request in 7)
    9 114.548085900     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=5/1280, ttl=64
   10 114.568330300     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=5/1280, ttl=63 (request in 9)
   11 115.550104796     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=6/1536, ttl=64
   12 115.570549596     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=6/1536, ttl=63 (request in 11)
   17 116.552283692     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=7/1792, ttl=64
   18 116.572923692     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=7/1792, ttl=63 (request in 17)
   19 117.554380889     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=8/2048, ttl=64
   20 117.574673389     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=8/2048, ttl=63 (request in 19)
   21 118.556152785     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=9/2304, ttl=64
   22 118.576669985     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=9/2304, ttl=63 (request in 21)
   23 119.558130881     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=10/2560, ttl=64
   24 119.578417081     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=10/2560, ttl=63 (request in 23)
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap2.pcap -Y icmp
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=1/256, ttl=64
    2 0.020429300     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=1/256, ttl=63 (request in 1)
    3 1.001061600     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=2/512, ttl=64
    4 2.027834200     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=3/768, ttl=64
    5 2.048083400     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=3/768, ttl=63 (request in 4)
    6 3.029705299     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=4/1024, ttl=64
    7 3.050185799     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=4/1024, ttl=63 (request in 6)
    8 4.032028597     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=5/1280, ttl=64
    9 4.052252897     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=5/1280, ttl=63 (request in 8)
   10 5.034047093     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=6/1536, ttl=64
   11 5.054471293     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=6/1536, ttl=63 (request in 10)
   16 6.036224789     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=7/1792, ttl=64
   17 6.056845489     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=7/1792, ttl=63 (request in 16)
   18 7.038326586     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=8/2048, ttl=64
   19 7.058596386     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=8/2048, ttl=63 (request in 18)
   20 8.040097082     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=9/2304, ttl=64
   21 8.060592082     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=9/2304, ttl=63 (request in 20)
   22 9.042072878     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=10/2560, ttl=64
   23 9.062339978     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=10/2560, ttl=63 (request in 22)
nishanth@LAPTOP-6Q3ONO7H:~$ sudo tshark -r /tmp/cap3.pcap -Y icmp
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=1/256, ttl=63
    2 0.000293700     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=1/256, ttl=64 (request in 1)
    3 2.027873300     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=3/768, ttl=63
    4 2.027948200     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=3/768, ttl=64 (request in 3)
    5 3.029927899     10.0.1.1 → 10.0.2.2     ICMP 98 Echo (ping) request  id=0x1b33, seq=4/1024, ttl=63
    6 3.030027799     10.0.2.2 → 10.0.1.1     ICMP 98 Echo (ping) reply    id=0x1b33, seq=4/1024, ttl=64 (request in 5)
```

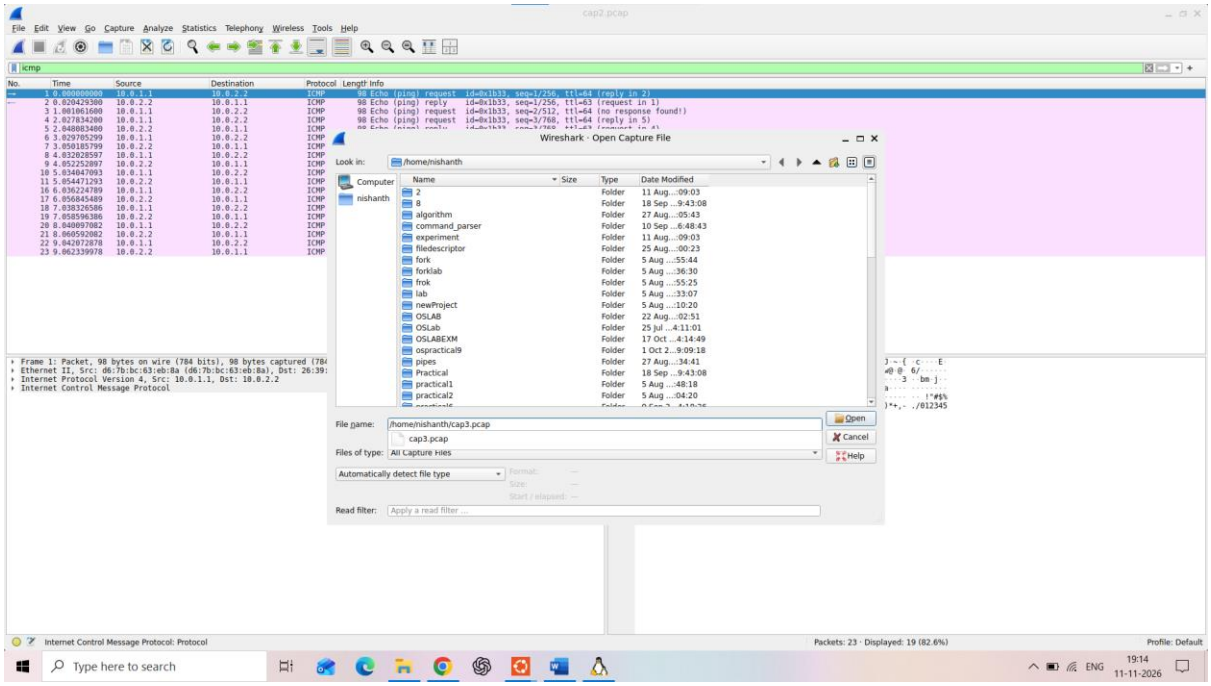
Open the captures in Wireshark:

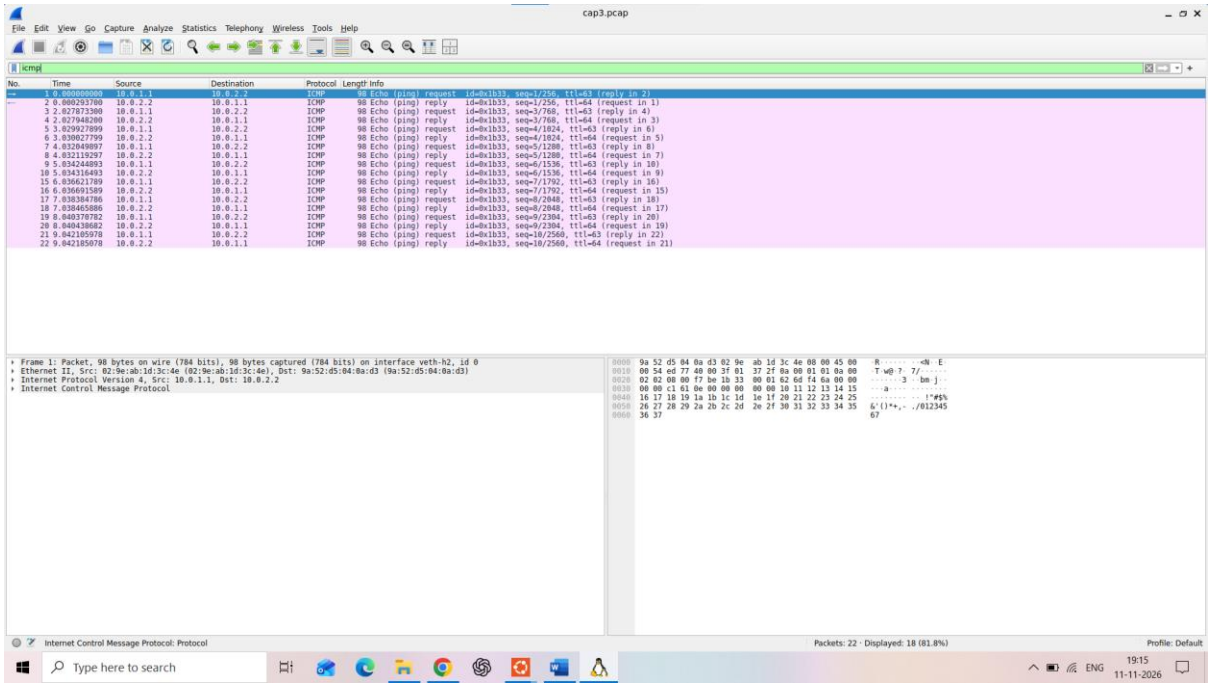


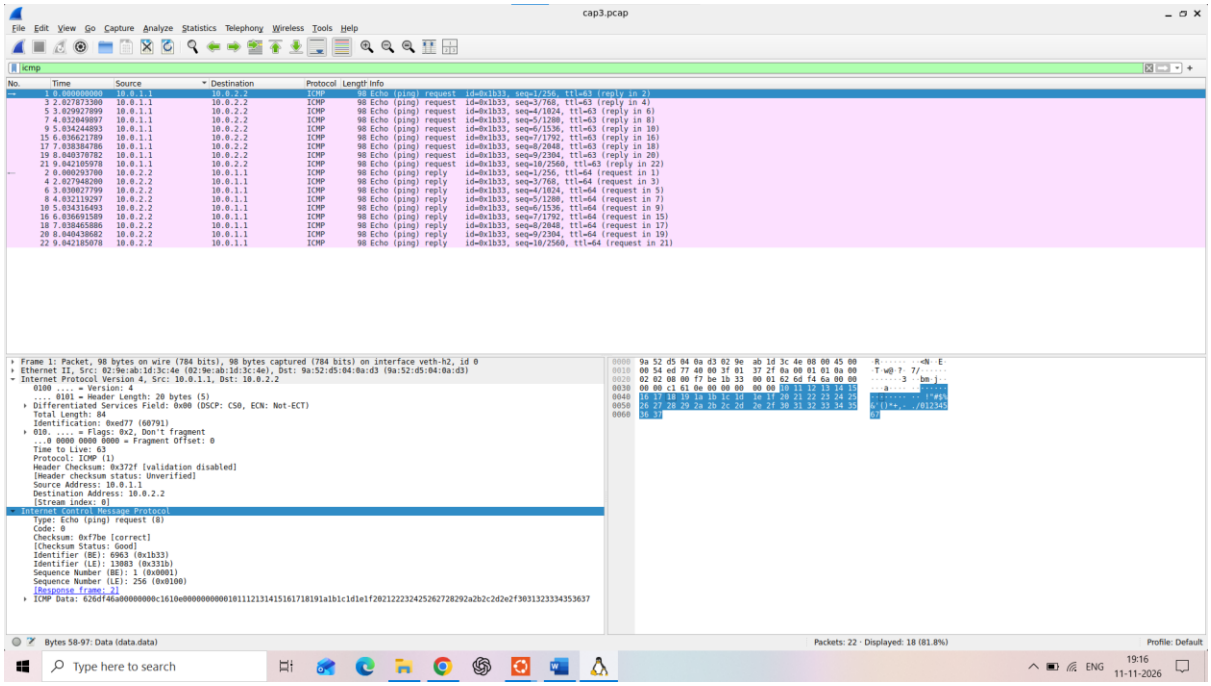

icmp
icmpv6
No. Source Destination Protocol Length Info
1 16058103 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=1/256, ttl=64 (reply in 3)
3 110.536506503 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=1/256, ttl=63 (request in 2)
4 111.517119503 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=2/512, ttl=64 (no response found!)
5 112.543809603 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=3/768, ttl=64 (reply in 6)
6 112.564161203 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=3/768, ttl=63 (request in 5)
7 113.545764202 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=4/1024, ttl=64 (reply in 8)
8 113.566264002 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=4/1024, ttl=63 (request in 7)
9 114.548085900 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=5/1280, ttl=64 (reply in 10)
10 114.568330300 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=5/1280, ttl=63 (request in 9)
11 115.550104796 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=6/1536, ttl=64 (reply in 12)
12 115.570549596 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=6/1536, ttl=63 (request in 11)
17 116.552283692 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=7/1792, ttl=64 (reply in 18)
18 116.572923692 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=7/1792, ttl=63 (request in 17)
19 117.554380889 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=8/2048, ttl=64 (reply in 20)
20 117.574673389 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=8/2048, ttl=63 (request in 19)
21 118.556152785 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=9/2304, ttl=64 (reply in 22)
22 118.576669985 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=9/2304, ttl=63 (request in 21)
23 119.558130881 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=10/2560, ttl=64 (reply in 24)
24 119.578417081 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=10/2560, ttl=63 (request in 23)
Frame 2: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface veth-h1, id 0
Ethernet II, Src: d6:7b:bc:63:eb:8a (d6:7b:bc:63:eb:8a), Dst: 26:39:a8:4a:bf:7e (26:39:a8:4a:bf:7e)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
Internet Control Message Protocol
0000 26 39 a8 4a bf 7e d6 7b bc 63 eb 8a 08 00 45 00 &9·J·~{ ·c····E·
0010 00 54 ed 77 40 00 40 01 36 2f 0a 00 01 01 0a 00 ·T·w@·@· 6/·····
0020 02 02 08 00 f7 be 1b 33 00 01 62 6d f4 6a 00 00 ·······3 ··bm·j··
0030 00 00 c1 61 0e 00 00 00 00 00 10 11 12 13 14 15 ···a···· ········
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 ········ ·· !"#$%
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()*+,- ./012345
0060 36 37 67


Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=1/256, ttl=64 (reply in 2)
2 0.020429300 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=1/256, ttl=63 (request in 1)
3 1.001061600 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=2/512, ttl=64 (no response found!)
4 2.027834200 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=3/768, ttl=64 (reply in 5)
5 2.048083400 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=3/768, ttl=63 (request in 4)
6 3.029705299 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=4/1024, ttl=64 (reply in 7)
7 3.050185799 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=4/1024, ttl=63 (request in 6)
8 4.032028597 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=5/1280, ttl=64 (reply in 9)
9 4.052252897 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=5/1280, ttl=63 (request in 8)
10 5.034047093 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=6/1536, ttl=64 (reply in 11)
11 5.054471293 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=6/1536, ttl=63 (request in 10)
12 5.195768893 26:39:a8:4a:bf:7e d6:7b:bc:63:eb:8a ARP 42 Who has 10.0.1.1? Tell 10.0.1.2
13 5.195846993 d6:7b:bc:63:eb:8a 26:39:a8:4a:bf:7e ARP 42 Who has 10.0.1.2? Tell 10.0.1.1
14 5.195941893 26:39:a8:4a:bf:7e d6:7b:bc:63:eb:8a ARP 42 10.0.1.2 is at 26:39:a8:4a:bf:7e
15 5.195909893 d6:7b:bc:63:eb:8a 26:39:a8:4a:bf:7e ARP 42 10.0.1.1 is at d6:7b:bc:63:eb:8a
16 6.036224789 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=7/1792, ttl=64 (reply in 17)
17 6.056845489 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=7/1792, ttl=63 (request in 16)
18 7.038326586 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=8/2048, ttl=64 (reply in 19)
19 7.058596386 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=8/2048, ttl=63 (request in 18)
20 8.040097082 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=9/2304, ttl=64 (reply in 21)
21 8.060592082 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=9/2304, ttl=63 (request in 20)
22 9.042072878 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=10/2560, ttl=64 (reply in 23)
23 9.062339978 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=10/2560, ttl=63 (request in 22)
Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface veth-r1, id 0
Ethernet II, Src: d6:7b:bc:63:eb:8a (d6:7b:bc:63:eb:8a), Dst: 26:39:a8:4a:bf:7e (26:39:a8:4a:bf:7e)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0000 26 39 a8 4a bf 7e d6 7b bc 63 eb 8a 08 00 45
0010 00 54 ed 77 40 00 40 01 36 2f 0a 00 01 01 0a
0020 02 02 08 00 f7 be 1b 33 00 01 62 6d f4 6a 00


icmp
icmpv6
No. Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=1/256, ttl=64 (reply in 2)
2 0.020429300 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=1/256, ttl=63 (request in 1)
3 1.001061600 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=2/512, ttl=64 (no response found!)
4 2.027834200 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=3/768, ttl=64 (reply in 5)
5 2.048083400 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=3/768, ttl=63 (request in 4)
6 3.029705299 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=4/1024, ttl=64 (reply in 7)
7 3.050185799 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=4/1024, ttl=63 (request in 6)
8 4.032028597 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=5/1280, ttl=64 (reply in 9)
9 4.052252897 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=5/1280, ttl=63 (request in 8)
10 5.034047093 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=6/1536, ttl=64 (reply in 11)
11 5.054471293 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=6/1536, ttl=63 (request in 10)
16 6.036224789 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=7/1792, ttl=64 (reply in 17)
17 6.056845489 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=7/1792, ttl=63 (request in 16)
18 7.038326586 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=8/2048, ttl=64 (reply in 19)
19 7.058596386 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=8/2048, ttl=63 (request in 18)
20 8.040097082 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=9/2304, ttl=64 (reply in 21)
21 8.060592082 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=9/2304, ttl=63 (request in 20)
22 9.042072878 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=10/2560, ttl=64 (reply in 23)
23 9.062339978 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=10/2560, ttl=63 (request in 22)
Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface veth-r1, id 0
Ethernet II, Src: d6:7b:bc:63:eb:8a (d6:7b:bc:63:eb:8a), Dst: 26:39:a8:4a:bf:7e (26:39:a8:4a:bf:7e)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
Internet Control Message Protocol
0000 26 39 a8 4a bf 7e d6 7b bc 63 eb 8a 08 00 45 00 &9·J·~{ ·c····E·
0010 00 54 ed 77 40 00 40 01 36 2f 0a 00 01 01 0a 00 ·T·w@·@· 6/·····
0020 02 02 08 00 f7 be 1b 33 00 01 62 6d f4 6a 00 00 ·······3 ··bm·j··
0030 00 00 c1 61 0e 00 00 00 00 00 10 11 12 13 14 15 ···a···· ········
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 ········ ·· !"#$%
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()*+,- ./012345
0060 36 37 67


cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
Wireshark · Open Capture File
Look in: /home/nishanth
Computer
nishanth
Name Size Type Date Modified
2 Folder 11 Aug...:09:03
8 Folder 18 Sep ...9:43:08
algorithm Folder 27 Aug...:05:43
command_parser Folder 10 Sep ...6:48:43
experiment Folder 11 Aug...:09:03
filedescriptor Folder 25 Aug...:00:23
fork Folder 5 Aug ...:55:44
forklab Folder 5 Aug ...:36:30
frok Folder 5 Aug ...:55:25
lab Folder 5 Aug ...:33:07
newProject Folder 5 Aug ...:10:20
OSLAB Folder 22 Aug...:02:51
OSLab Folder 25 Jul ...4:11:01
OSLABEXM Folder 17 Oct ...4:14:49
ospractical9 Folder 1 Oct 2...9:09:18
pipes Folder 27 Aug...:34:41
Practical Folder 18 Sep ...9:43:08
practical1 Folder 5 Aug ...:48:18
practical2 Folder 5 Aug ...:04:20
File name: /home/nishanth/cap3.pcap
cap3.pcap
Files of type: All Capture Files
Automatically detect file type
Format:
Size:
Start / elapsed:
Read filter: Apply a read filter ...
Open
Cancel
Help
Internet Control Message Protocol: Protocol
Packets: 23 · Displayed: 19 (82.6%)
Profile: Default
Type here to search
ENG
19:14
11-11-2026


cap3.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
Internet Control Message Protocol: Protocol
Packets: 22 · Displayed: 18 (81.8%)
Profile: Default
Type here to search
ENG
19:15
11-11-2026


cap3.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=1/256, ttl=63 (reply in 2)
3 2.027873300 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=3/768, ttl=63 (reply in 4)
5 3.029927899 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=4/1024, ttl=63 (reply in 6)
7 4.032049897 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=5/1280, ttl=63 (reply in 8)
9 5.034244893 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=6/1536, ttl=63 (reply in 10)
15 6.036621789 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=7/1792, ttl=63 (reply in 16)
17 7.038304786 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=8/2048, ttl=63 (reply in 18)
19 8.040370782 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=9/2304, ttl=63 (reply in 20)
21 9.042105978 10.0.1.1 10.0.2.2 ICMP 98 Echo (ping) request id=0x1b33, seq=10/2560, ttl=63 (reply in 22)
2 0.000293700 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=1/256, ttl=64 (request in 1)
4 2.027948200 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=3/768, ttl=64 (request in 3)
6 3.030027799 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=4/1024, ttl=64 (request in 5)
8 4.032119297 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=5/1280, ttl=64 (request in 7)
10 5.034316493 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=6/1536, ttl=64 (request in 9)
16 6.036691589 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=7/1792, ttl=64 (request in 15)
18 7.038465886 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=8/2048, ttl=64 (request in 17)
20 8.040438682 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=9/2304, ttl=64 (request in 19)
22 9.042185078 10.0.2.2 10.0.1.1 ICMP 98 Echo (ping) reply id=0x1b33, seq=10/2560, ttl=64 (request in 21)
Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface veth-h2, id 0
Ethernet II, Src: 02:9e:ab:1d:3c:4e (02:9e:ab:1d:3c:4e), Dst: 9a:52:d5:04:0a:d3 (9a:52:d5:04:0a:d3)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0xed77 (60791)
010. .... = Flags: 0x2, Don't fragment
..0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 63
Protocol: ICMP (1)
Header Checksum: 0x372f [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
[Stream index: 0]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0xf7be [correct]
[Checksum Status: Good]
Identifier (BE): 6963 (0x1b33)
Identifier (LE): 13083 (0x331b)
Sequence Number (BE): 1 (0x0001)
Sequence Number (LE): 256 (0x0100)
[Response frame: 2]
ICMP Data: 626df46a00000000c1610e0000000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637
0000 9a 52 d5 04 0a d3 02 9e ab 1d 3c 4e 08 00 45 00
0010 00 54 ed 77 40 00 3f 01 37 2f 0a 00 01 01 0a 00
0020 02 02 08 00 f7 be 1b 33 00 01 62 6d f4 6a 00 00
0030 00 00 c1 61 0e 00 00 00 00 00 10 11 12 13 14 15
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
0060 36 37
Bytes 58-97: Data (data.data)
Packets: 22 · Displayed: 18 (81.8%)
Profile: Default
Type here to search
ENG
19:16
11-11-2026